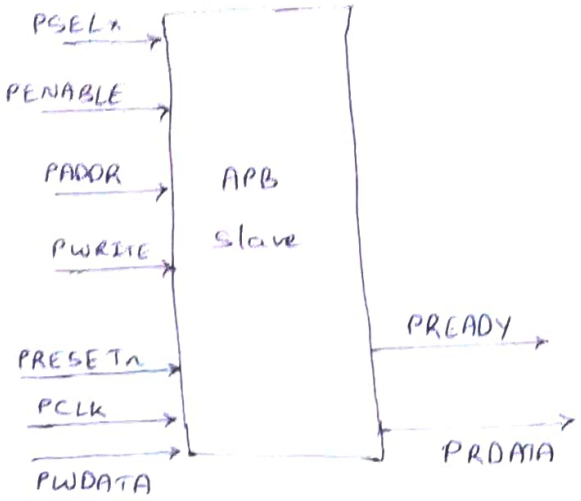
V-plan :-

Project Name :- APB based SPI Protocol

Amrit

* APB Block diagram :-



- PCLK (clock) : Rising edge synchronizes all signals
- PRESETn (Reset) : Active - low reset, often connected directly to system reset (Active - low signal);
- PSELx (Slave select) : Driven by the master to select for a target slave.
- PENABLE (Enable) : Indicates the second cycle (Access phase) of an APB transfer
- PWRITE (write / Read) : High indicates a write transfer, Low indicates a Read transfer
- PADDR (Address) : Address bus (3-bit signal)
- PWDATA (write data) : Valid during write operations (8-bit signal)
- PRDATA (Read data) : Driven by the selected slave during read operations (8-bit signal)
- PREADY (Ready) : Driven by the slave to extend the transfer if its busy
- PSLVERR (slave Error) : Optional slave signal indicating transfer.

SPI Block diagram :-



- SCLK (serial Clock) : Generated by the master this signal synchronizes data transmission. One bit is transferred per clock pulse
- MOSI (Master Out slave in) : Data line used by the master to send data to the slave device
- MISO (Master In slave out) : Data line used by the slave to send data back to the master.
- SS (slave select) : An active low signal used by the master to select specific slaves allowing multiple devices to share the same bus.

Objectives of project :-

Following are the objectives of project :-

① Verification of APB - SPI Interoperability :-
Confirming that the APB slave interface correctly decodes addresses (PADDR) and transfers data (PWDATA / PRDATA) from the master bus to the internal SPI registers

② Functional Accuracy :-
Verifying all SPI modes, including clock polarity / clock phase (CPOL / CPHA) baud rate generation, MSB / LSB first transmission and multi-slave select

③ Verification of Low power conservation mode :-
when the SPI is in "stop mode" the SCLK is not generated we are verifying.

④ Verification of all the 8 modes of operation of CPOL / CPHA by considering LSBFE = 1 and LSBFE = 0.

APB features :-

1. It is a synchronous protocol uses PCLK for synchronization
2. To communicate between high frequency component and low frequency peripherals.
3. It is a low power, low bandwidth, simple interface
4. Non-pipeline - At a time one transaction can be sent
5. No burst transfer
6. Simple handshake mechanism by the help of PREADY
7. It uses 10 signals for its communication.

SPI features :-

1. Master mode support only - The master can initiate data transfer and make ss signal low and start the data transfer (support single Master and multiple slave configuration).
2. Bi-directional mode - which is a full duplex mode at a time the master can send data (MOSI) and receive data (MISO).
3. Mode fault error flag with CPU interrupt capability - Depends on the SSOE bit present in control register-1 so the ss signal is controlled by hardware (DUT) / software (TB).
   - if SSOE = 0 (ss low manually) - by test bench
   - if SSOE = 1 (ss low by hardware) - by DUT
4. Double-buffered data register :- The registers present in SPI can send data and receive data in a single clock.
5. Serial Clock with Programmable Polarity & phase - The CPOL and CPHA bits present in the control register-1 provides the alignment of data depending on the modes

mode	CPOL	CPHA
0	0	0
1	0	1
2	1	0
3	1	1

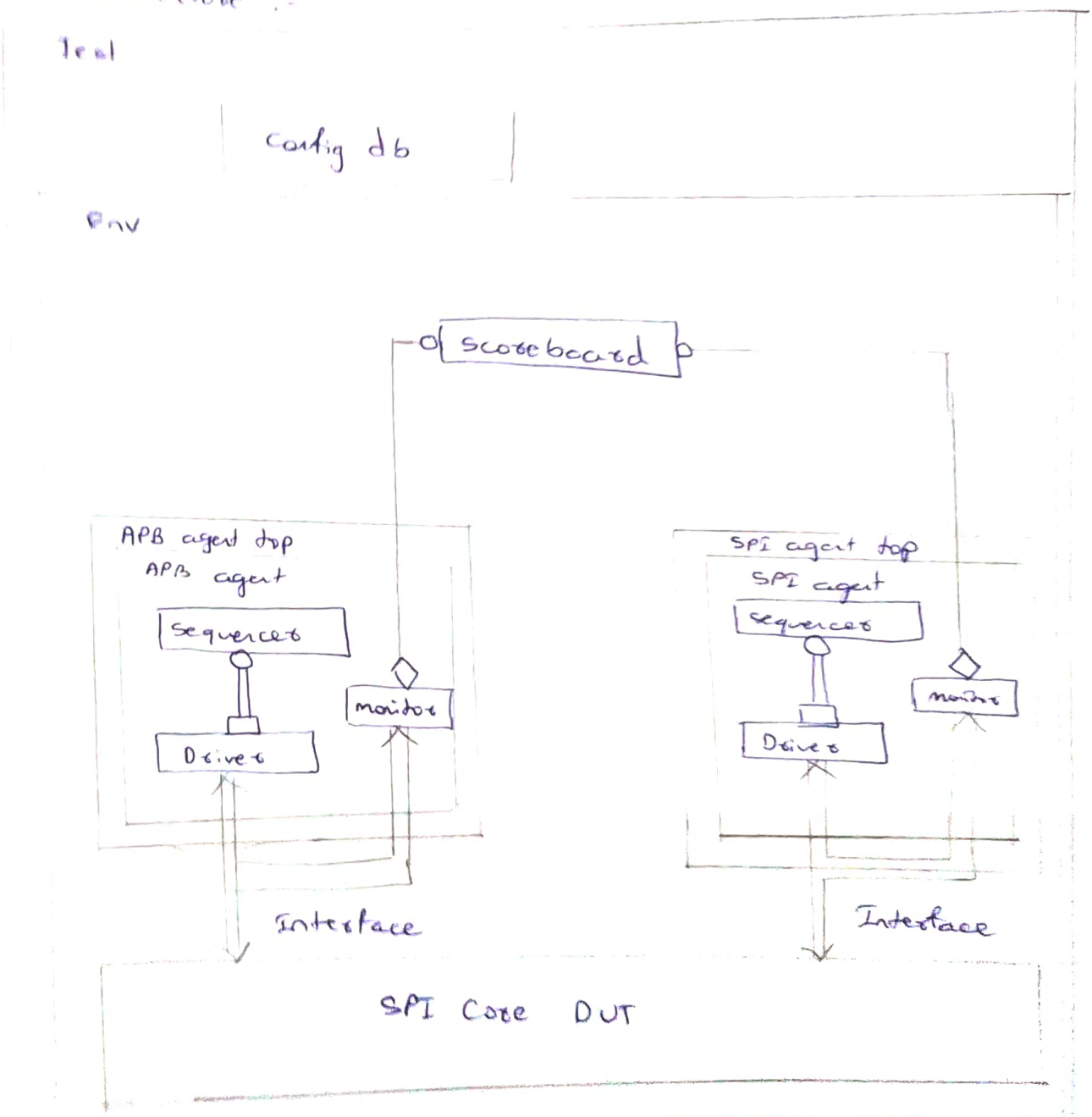
(6) Control of SPI operation during wait mode :- The SPI operates on 3 - modes (Run mode, wait mode, stop mode) which is configured by the SPE bit and SPISWAI bits present in control register 1 and control register - 2.

(1) RUN MODE (SPE = 1)

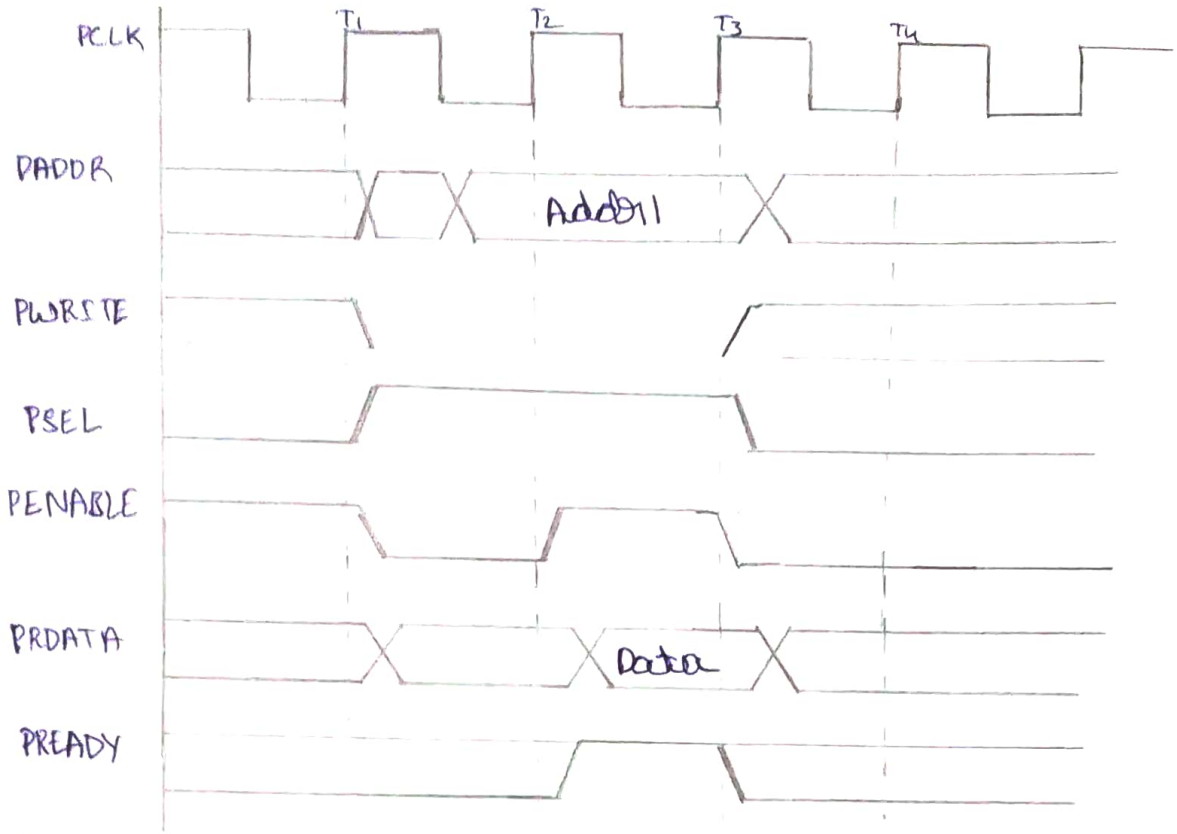
(2) WAIT MODE (SPISWAI = 0, SPE = 0)

(3) STOP MODE (SPISWAI = 1, SPE = 0)

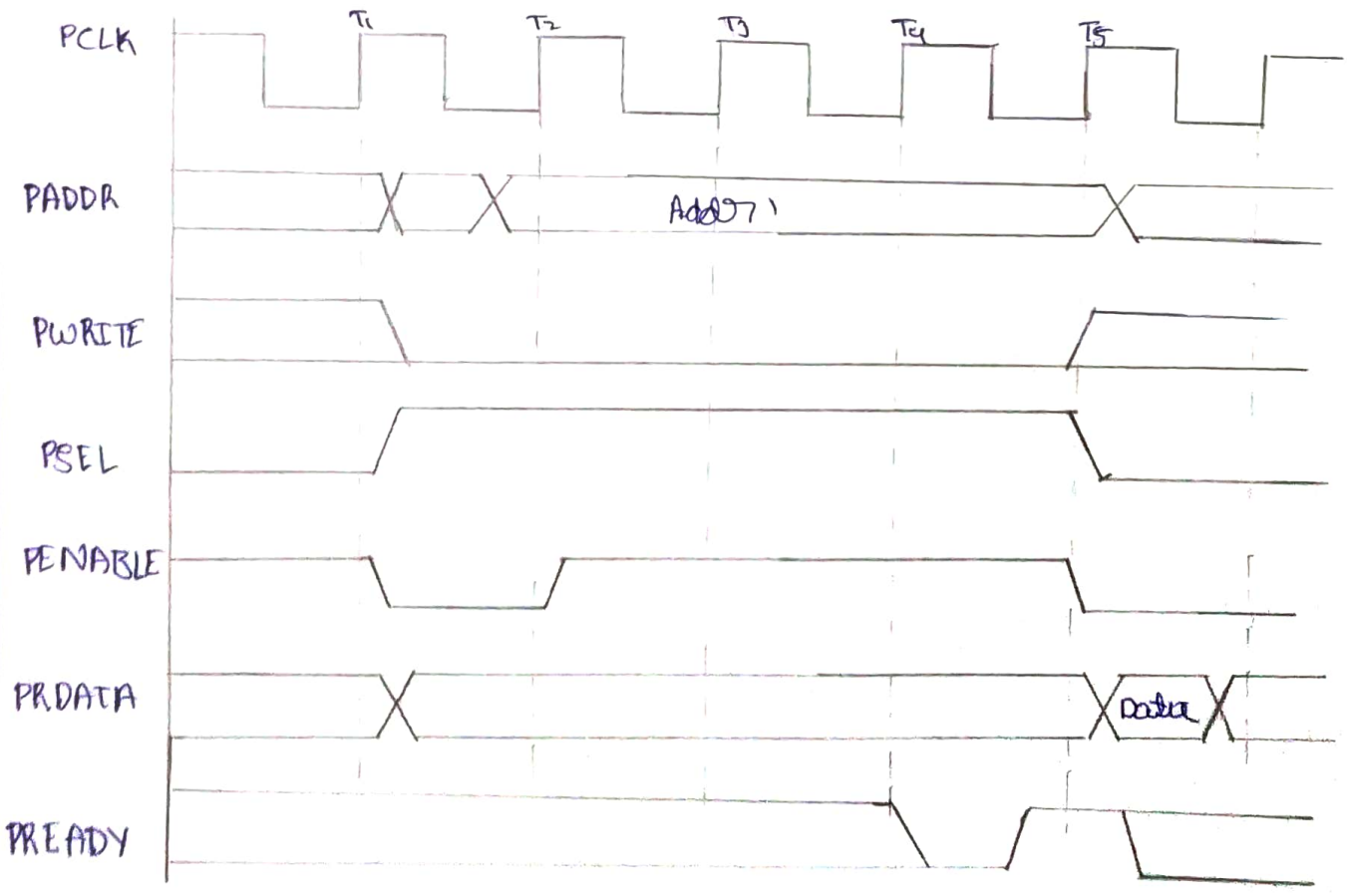
Test Bench Architecture :-



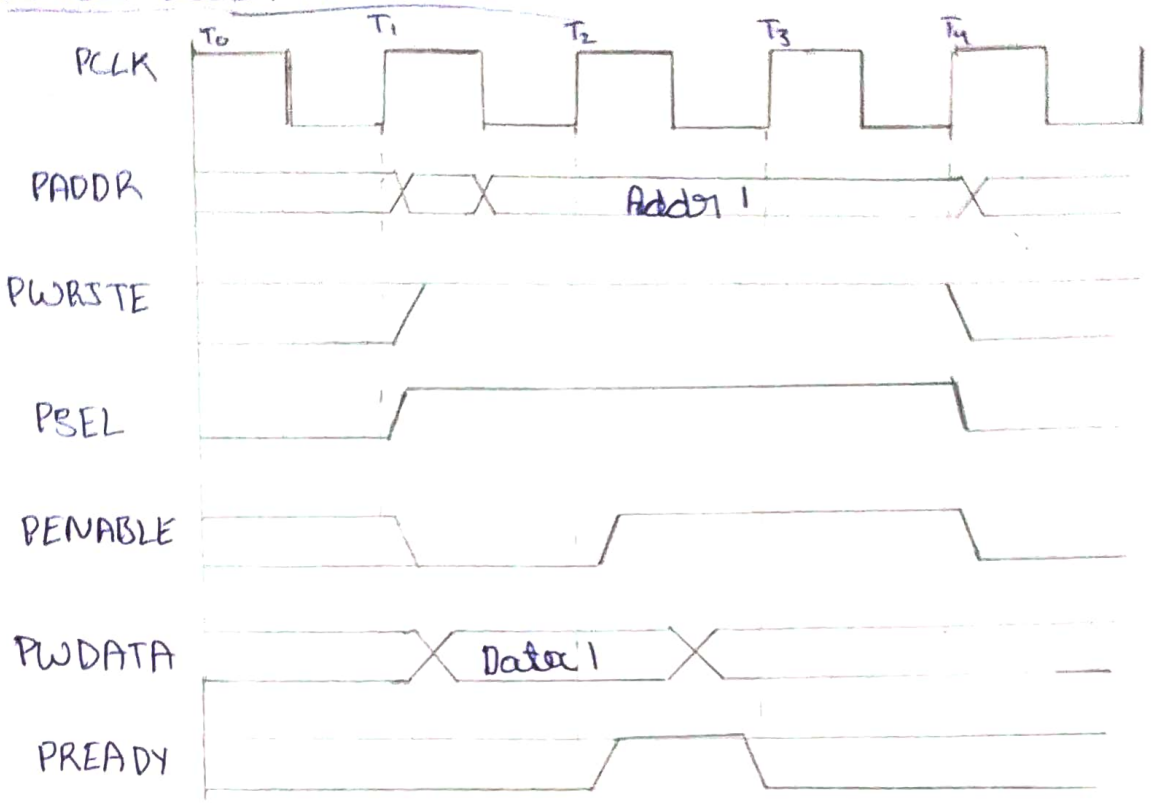
Read with no wait State:



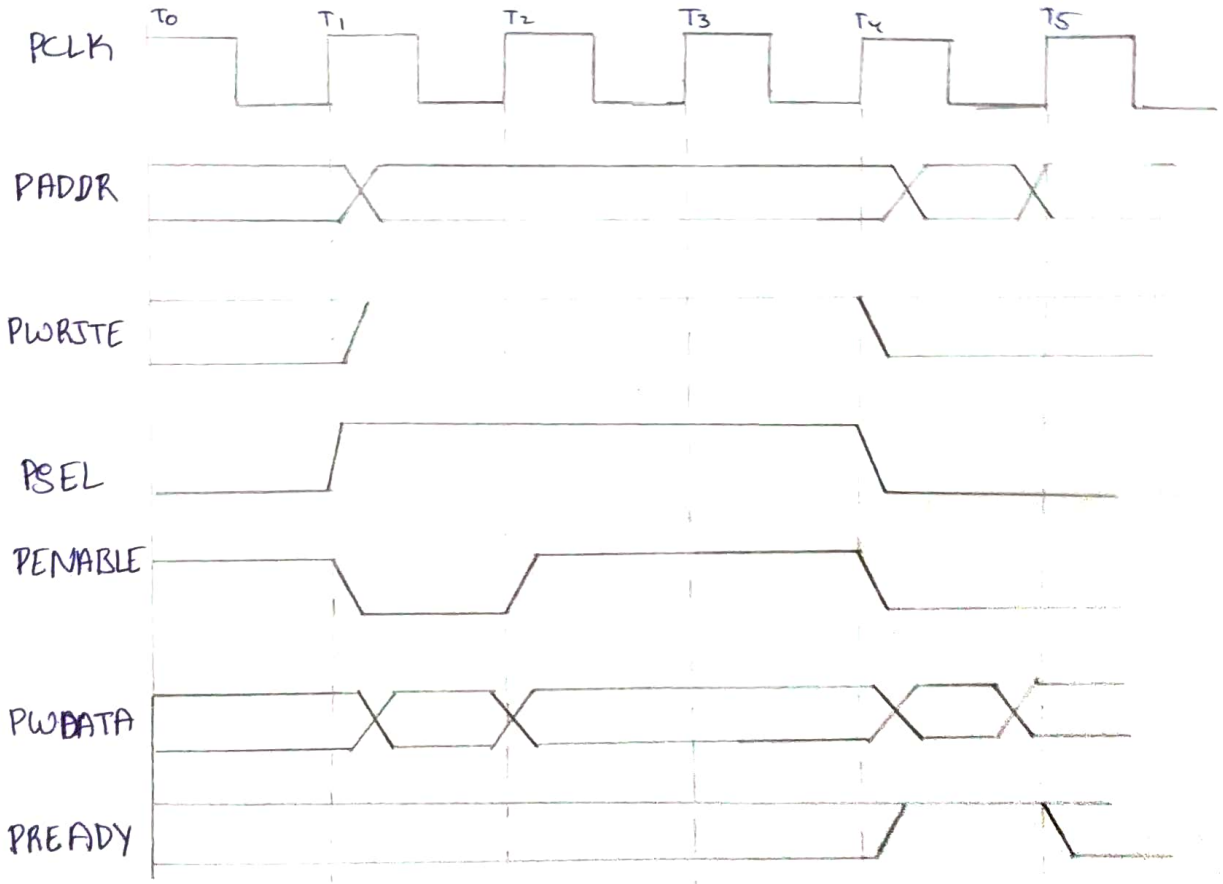
Read with wait State



write with no wait



write with wait state.



Scoreboard :-

1. Capture Expected data :-
   Inside scoreboard, it will capture the data which APB master sent.
2. Capture Actual data :-
   → Inside scoreboard, it will capture the data which is the actual output generated by the SPI
   → APB will read PRDATA available
3. Compare Expected with actual data :-
   Scoreboard will check each bit by bit if the actual output is equal to the expected output then scoreboard will print successful.

APB monitor → Expected data
SPI monitor → Actual data } Scoreboard (compares)

Process :-

→ APB starts the communication by writting the data through PWDATA to the SPI_DR - parallely
→ Inside the SPI, shift register stores the data and converts it into serial and sends one by one to the slave through MOSI.
→ Then slave will send serial data through MISO to the SPI that data is available in the data register after 8th clock cycle.
→ Inside SPI shift register will convert serial data to parallel data that data APB will take as PRDATA
→ Whatever SPI sent the data actual data (PRDATA) and Expected data from APB will be compared.

Testcases :-

Total 8 - test cases are verified :-

① CPOL = 0, CPHA = 0, LSBFE = 0

② CPOL = 0, CPHA = 1, LSBFE = 0

③ CPOL = 1, CPHA = 0, LSBFE = 0

④ CPOL = 1, CPHA = 1, LSBFE = 0

⑤ CPOL = 0, CPHA = 0, LSBFE = 1

⑥ CPOL = 0, CPHA = 1, LSBFE = 1

⑦ CPOL = 1, CPHA = 0, LSBFE = 1

⑧ CPOL = 1, CPHA = 1, LSBFE = 1

⑨ Low power consumption mode (STOP MODE)
SPISWAI = 1, SPE = 0

Functional Coverage :-

① Covergroup for APB_signals

② Covergroup for SPI signals.

Block diagram :-

